

1(a)	units of F_D : kg m s^{-2}	M1
	units of ρ : kg m^{-3} and units of A : m^2 and units of v : m s^{-1} or units of v^2 : $\text{m}^2 \text{s}^{-2}$	M1
	$\text{kg m s}^{-2} = C \text{ kg m s}^{-2}$ and comment '(so) C has no units' / unit terms cancelled or $C = \text{kg m s}^{-2} / (\text{kg m}^{-3} \text{m}^2 \text{m}^2 \text{s}^{-2})$ and comment '(so) C has no units' / unit terms cancelled	A1
1(b)	one arrow vertically downward labelled weight to within 10° of the vertical	B1
	one arrow vertically upwards labelled drag / drag force / F_D / air resistance / viscous force to within 10° of the vertical	B1
1(c)	(at terminal velocity) $F_D = mg$	C1
	$F_D = 0.049 \times 9.81$ $= 0.48 \text{ N}$	A1
1(d)	area = $\pi \times (0.060 / 2)^2$	C1
	$0.48 = \frac{1}{2} \times C \times 1.2 \times \pi \times (0.060 / 2)^2 \times 25^2$	C1
	$C = 0.45$	A1